

Fact Sheet: East Brandenburg

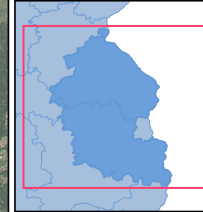
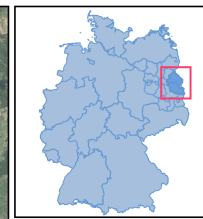
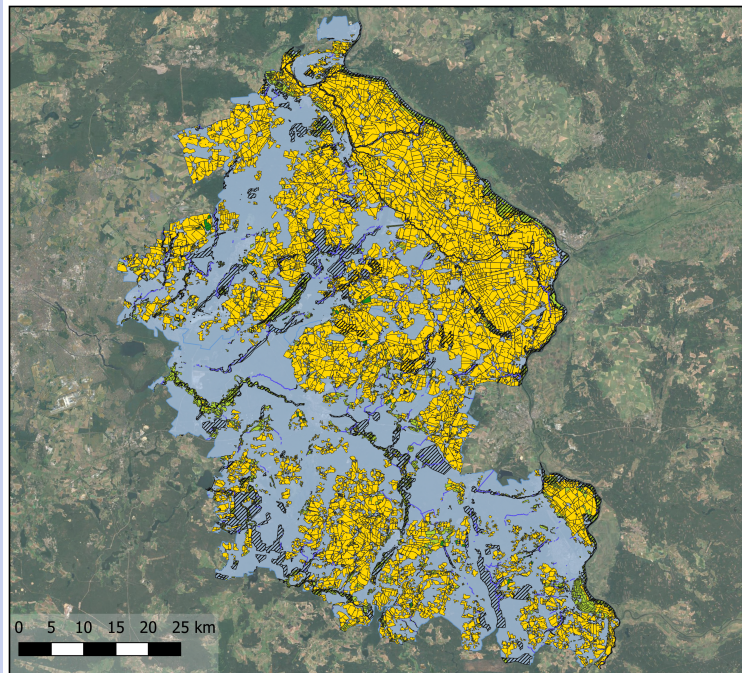
Name of living lab

Contact: Julia Gunnoltz (julia.gunnoltz@zalf.de)

Name and email



Please add GIS map here



Natura 2000 Area
 Arable land
 Permanent cultures
 Grass land
 Streams

Soil-climate area (Bodenklimaräume)

Please describe the main characteristics regarding the geo-physical conditions from perspective of agriculture (Soil types and conditions for agriculture; Climate conditions; Geographic characteristics)

Geographic characteristics:

Lowland region shaped by multiple Ice Age glaciations

Minimal elevation differences with predominantly flat terrain with Oderbruch as a wide river valley shaped by glacial meltwater

Soil Types and conditions for agriculture:

Dominated by sandy and loamy-sandy soils ("Märkischer Sand") with low natural fertility

Müncheberger Soil Rating: Mostly very low to low soil quality (below 50 points);

medium-quality soils only in specific floodplain regions (e.g., Oderbruch)

High spatial heterogeneity of soil quality within single fields due to glacial landscape formation

Climate conditions:

Continuous warming trend observed since 1881: +1.3 °C increase in mean annual temperature to date

Short-term climate projections (2021–2050): Additional warming of +1.1 to +1.5 °C

Shift in climate extremes: More hot summer days, Fewer frost days, hail and storm, spring drought

increases expected in eastern Brandenburg due to stronger continental influence

Key challenges (Please describe the key challenges related to soil climate area)

High drought vulnerability due to limited soil water retention capability

Greater irrigation demand and competition for scarce water resources

Land degradation risks, including organic matter loss and declining soil health

Biodiversity decline in agricultural landscapes (e.g., drying of breeding habitats for amphibians)

Soil-climate information

Main soil types:	sandy - loamy-sandy
Topography:	mostly flat
Site altitude (m):	50 m (average)
Average rainfall (mm/year):	558 (1960-90)
Average temperature:	9,2 °C (BB)

Justification for living lab delineation

Marginal, intensively farmed agricultural region with low ecological and economic diversity facing climate challenges, suitable for testing climate adaptation strategies and coordinated agri-food system interventions

Agricultural production system
Please describe the main characteristics regarding agricultural production systems: Total agricultural area: 123,600 ha; 114,100 ha arable land (MOL) Main crops: winter grain (>50%), maize for silage/energy (~16,800 ha), winter rapeseed (~14,100 ha), Other land uses: horticulture and seed/plant cultivation minimal; permanent grassland ~9,300 ha. Animal husbandry: poultry most common; pigs, cattle, sheep, goats declining Land ownership: much farmland not owned by farmers climate stresses affect production Region is becoming a hub for renewable energy, Land-use conflict: efficiency measures in agriculture vs. energy expansion
Key challenges (Please describe the key challenges related to the agricultural production system) Reduced crop productivity and stability under intensifying climate stress High adaptation pressure on farmers, requiring new practices and investments Competition for land between agriculture and renewable energy expansion Decline in livestock diversity (pigs, cattle, sheep, goats) Limited horticulture and specialized crop production (low diversification) Balancing organic and conventional farming practices while maintaining productivity

General information	
Land Area:	4424,89 km ²
Agriculture:	1757,11 km ²
Forests:	1624,34 km ²
Grassland:	243,53 km ²
Fallow land:	104,88 km ²
Protected area:	2.351,52 km ²
Organic agriculture:	5,3 % (2018)
Main crop types:	winter grain (rye, wheat), maize, winter rapeseed
Main livestock:	poultry (chicken), bees
Average farm size:	242 ha/ farm
Average rent rate:	€302 / €152

Socio-economic conditions
Please describe the main characteristics regarding the socio-cultural conditions: Higher population density and employment near Berlin; commuter flows mainly from eastern border toward Berlin, Low population density, aging population, and outmigration of young people reduce workforce availability. Employment structure: Predominantly services (tourism, healthcare, social services), followed by manufacturing/trades; agriculture employs few despite large land areas Economic hubs: Tesla Gigafactory and Berlin-Brandenburg Airport (LOS) attract businesses; Berlin metro area provides market opportunities. Growth trends: Commercial development and employment expected to increase; LOS GDP (€38,672) above MOL (€25,177), LOS shows 22% growth. Tourism: Driven by nature parks, protected areas, and spa towns.
Key challenges (Please describe the key challenges related to socio-cultural conditions) Agriculture and peripheral industries generate limited regional value creation Weak local processing and value-adding facilities limits regional economic integration, many areas faced a decline of traditional manufacturing and industrial sites Skilled labor shortage (Fachkräftemangel) compounded by competition with Berlin's dynamic job market

General information	
Population: (inhabitants)	72,275
Population density: (inhabitants/km ²):	40/ km ²
Next city:	Berlin
GDP/Capita:	25,177 38,672
Average annual household income: (€)	€ 24,466 Disposable income per capita (2022)
NUTS 3 codes:	DE409-40C

Key stakeholders (organisations / institutions)
Local Leader Groups (LAG), regional professional associations including farmers associations, regional municipalities, nature conservation associations, nature park, business support service, pioneers, private initiatives, adult education institutions

Ongoing research (topics)
Mixed Cropping Systems (Patchcrop Field Experiment) Large-Seeded Grain Legumes and Their Value Chains Agri-PV - Assessment and Acceptance Soil Health - Preservation and Improvement, Analysis of the Effects of Measures (such as agroforestry, robotics, reduced tillage) Cooperative Agri-Environmental Measures